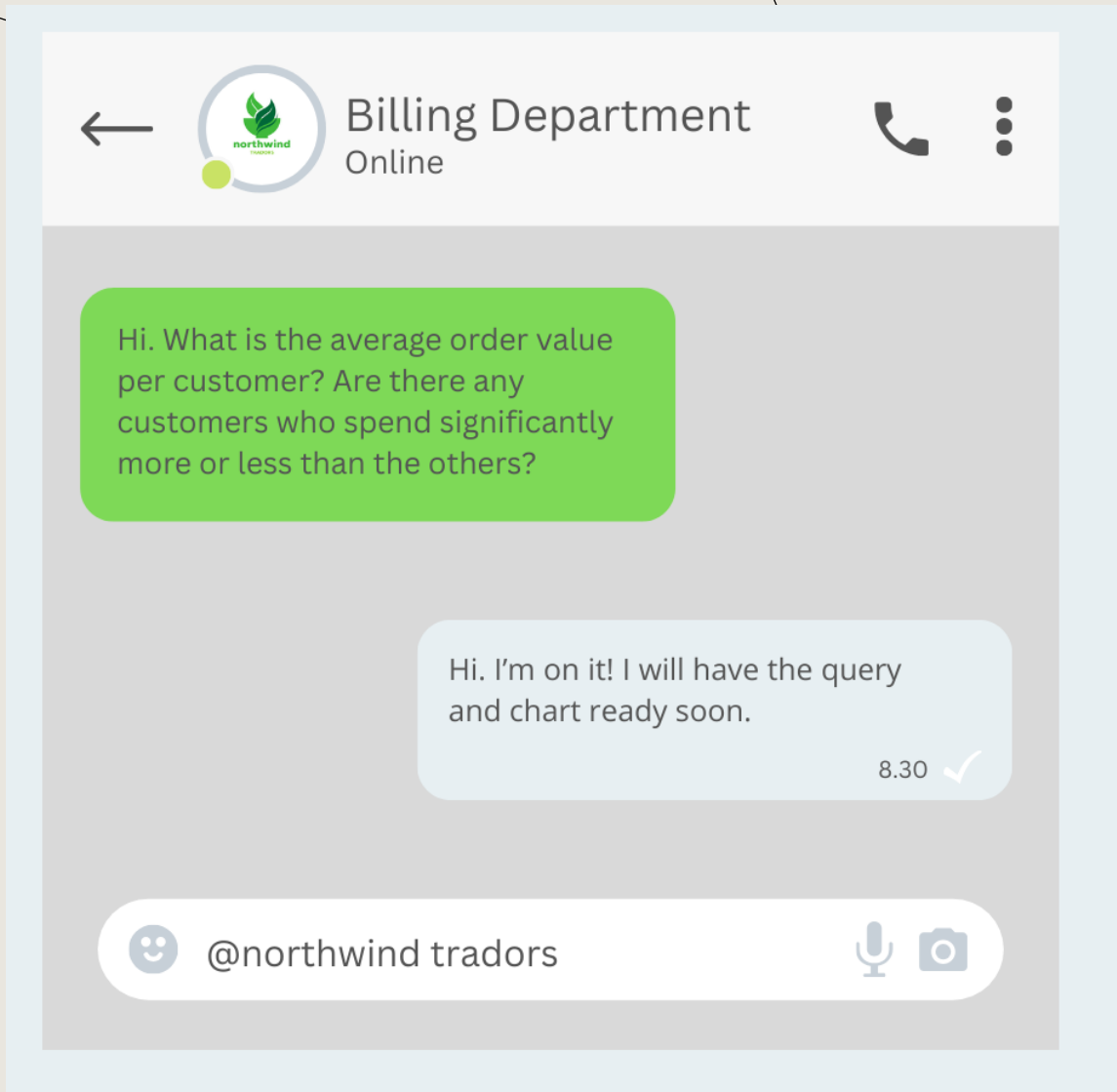


SQL DATABASE PROJECT





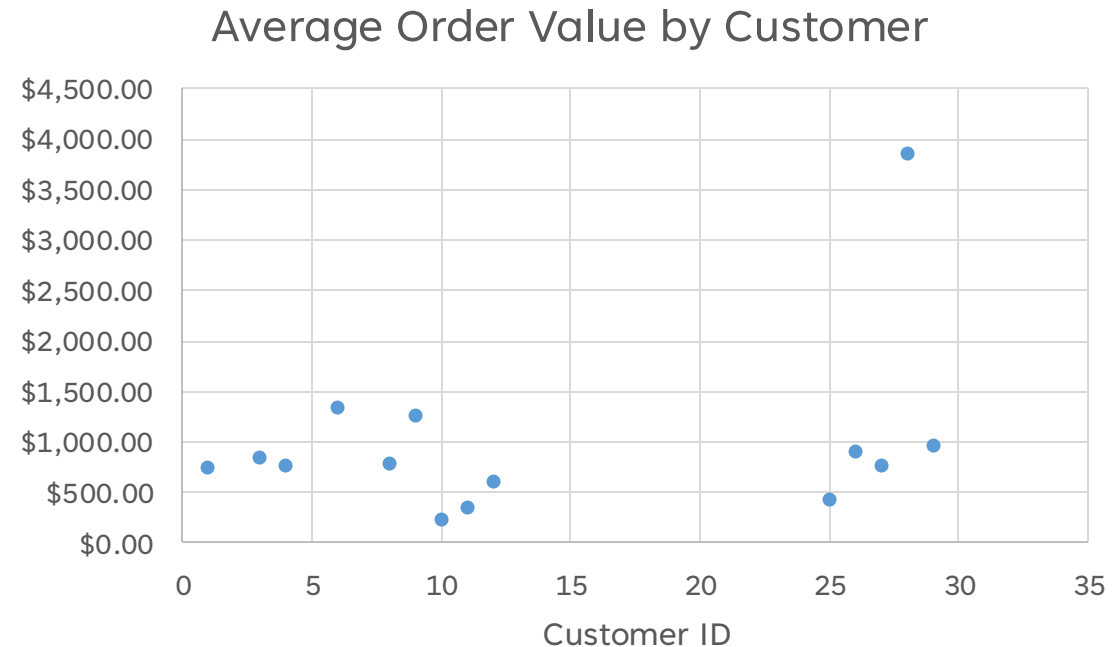
CUSTOMER SPENDING:

The Billing Department wants to know: What is the average order value per customer? Are there any customers who spend significantly more or less than the others?

Average Order Value by Customer

```
select orders.Customer, ROUND(avg(`Unit Price` * `Quantity`),2) As `Average Price`  
from `order details`  
JOIN orders ON `order details`.`Order ID` = orders.`Order ID`  
GROUP BY orders.Customer  
order by Customer asc;
```

	Customer	Average Price
▶	1	736
	3	850
	4	761.5
	6	1334.58
	8	780.5
	9	1262.17
	10	235.42
	11	339.83
	12	595
	25	430
	26	906.31
	27	752.5
	28	3858.12
	29	968.5

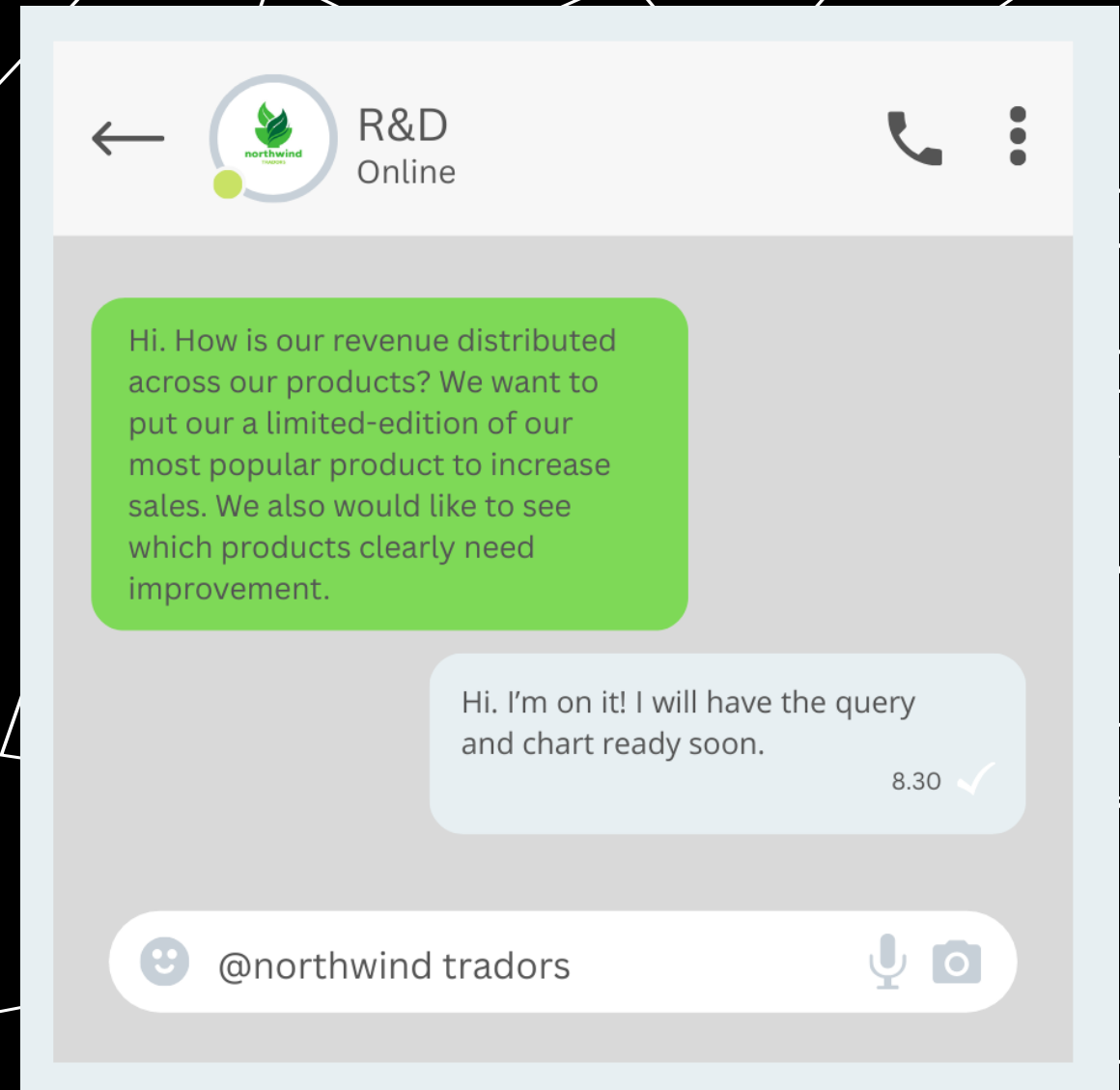


I send this table and scatterplot to the Billing Department. They can clearly see that our customers usually spend between \$500 and \$1000. They can also see that Customer 28 is our biggest client.

PRODUCT POPULARITY:

The R&D Team would like to know:

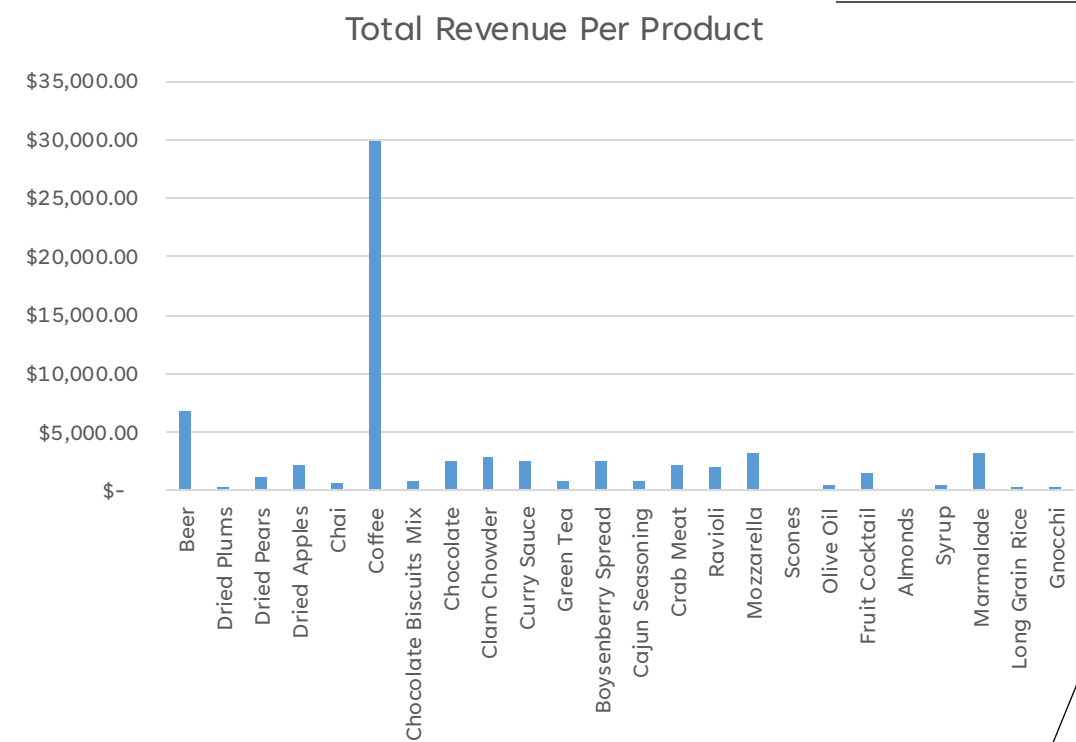
How is our revenue distributed across our products? We want to put out a limited-edition of our most popular product to increase sales. We also would like to see which products clearly need improvement.



Total Revenue per Product

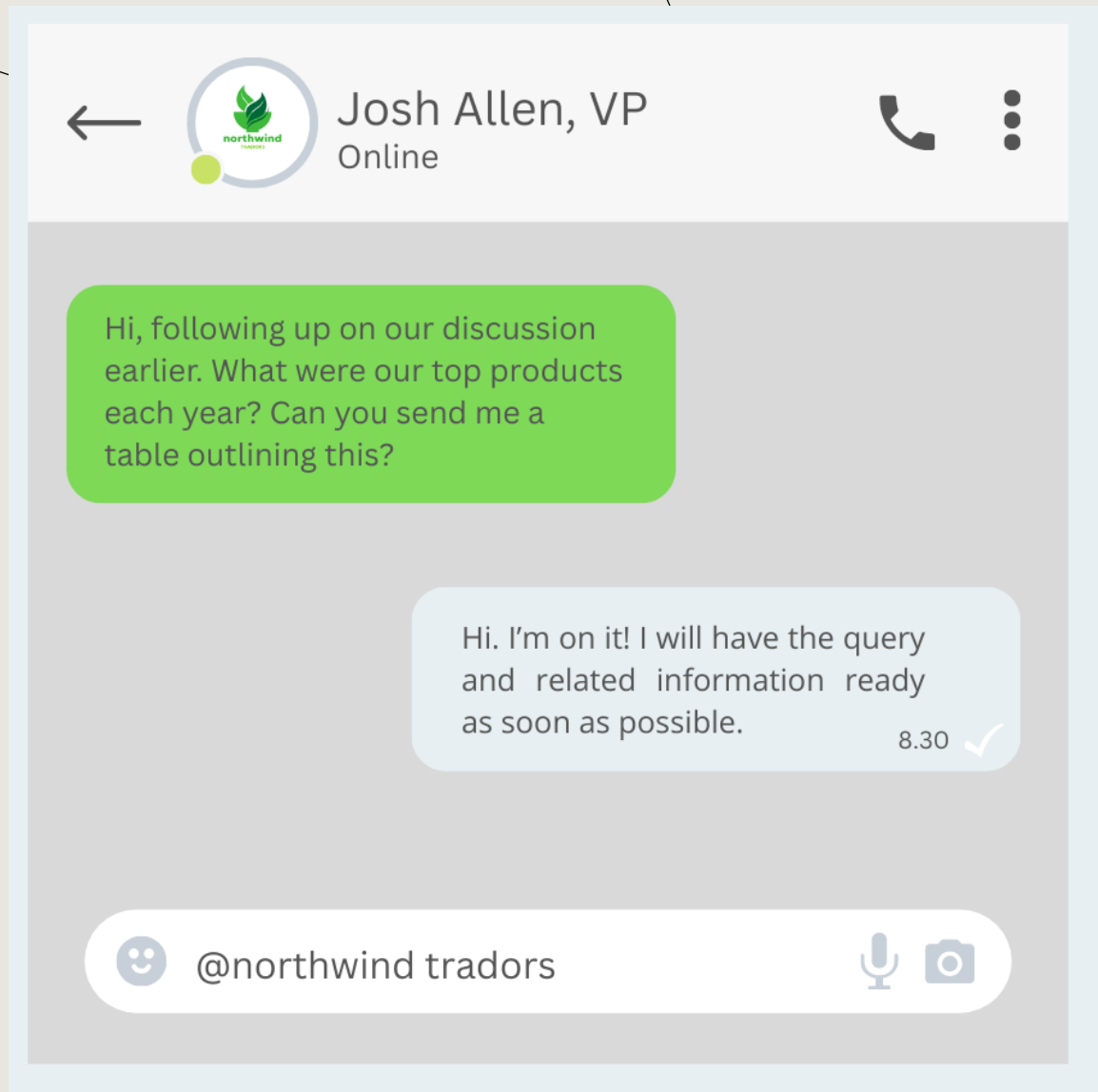
	Total Revenue	Product
▶	6818	Northwind Traders Beer
	262.5	Northwind Traders Dried Plums
	1200	Northwind Traders Dried Pears
	2120	Northwind Traders Dried Apples
	720	Northwind Traders Chai
	29900	Northwind Traders Coffee
	782	Northwind Traders Chocolate Biscuits Mix
	2550	Northwind Traders Chocolate
	2798.5	Northwind Traders Clam Chowder
	2600	Northwind Traders Curry Sauce
	822.25	Northwind Traders Green Tea
	2500	Northwind Traders Boysenberry Spread
	880	Northwind Traders Cajun Seasoning
	2208	Northwind Traders Crab Meat
	1950	Northwind Traders Ravioli
	3132	Northwind Traders Mozzarella
	200	Northwind Traders Scones
	533.75	Northwind Traders Olive Oil
	1560	Northwind Traders Fruit Cocktail
	200	Northwind Traders Almonds
	500	Northwind Traders Syrup
	3240	Northwind Traders Marmalade
	280	Northwind Traders Long Grain Rice
	380	Northwind Traders Gnocchi

```
select sum(`Unit Price` * `Quantity`) as `Total Revenue`, Product
from `order details`
Group by Product;
```



I send this bar chat to R&D so they can clearly see how our profit is distributed among our products. They go right to work on producing a limited-edition coffee flavor, clearly our most popular product.





TOP PRODUCTS:

The VP wants to know: What were our top products each year? Can you send me a table outlining this?

View of Top Products per Year

```
CREATE VIEW `TopSellingProductsPerYear` AS
SELECT
  p.`Product Name`,
  p.`ID` AS `Product ID`,
  YEAR(STR_TO_DATE(o.`Order Date`, '%m/%d/%Y')) AS `Year`,
  SUM(od.`Quantity` * od.`Unit Price`) AS `Total Sales`
FROM
  `order details` od
JOIN
  `orders` o ON od.`Order ID` = o.`Order ID`
JOIN
  `products` p ON od.`Product` = p.`Product Name`
GROUP BY
  p.`Product Name`, p.`ID`, YEAR(STR_TO_DATE(o.`Order Date`, '%m/%d/%Y'));

SELECT *
FROM TopSellingProductsPerYear
ORDER BY `Total Sales` DESC;
```

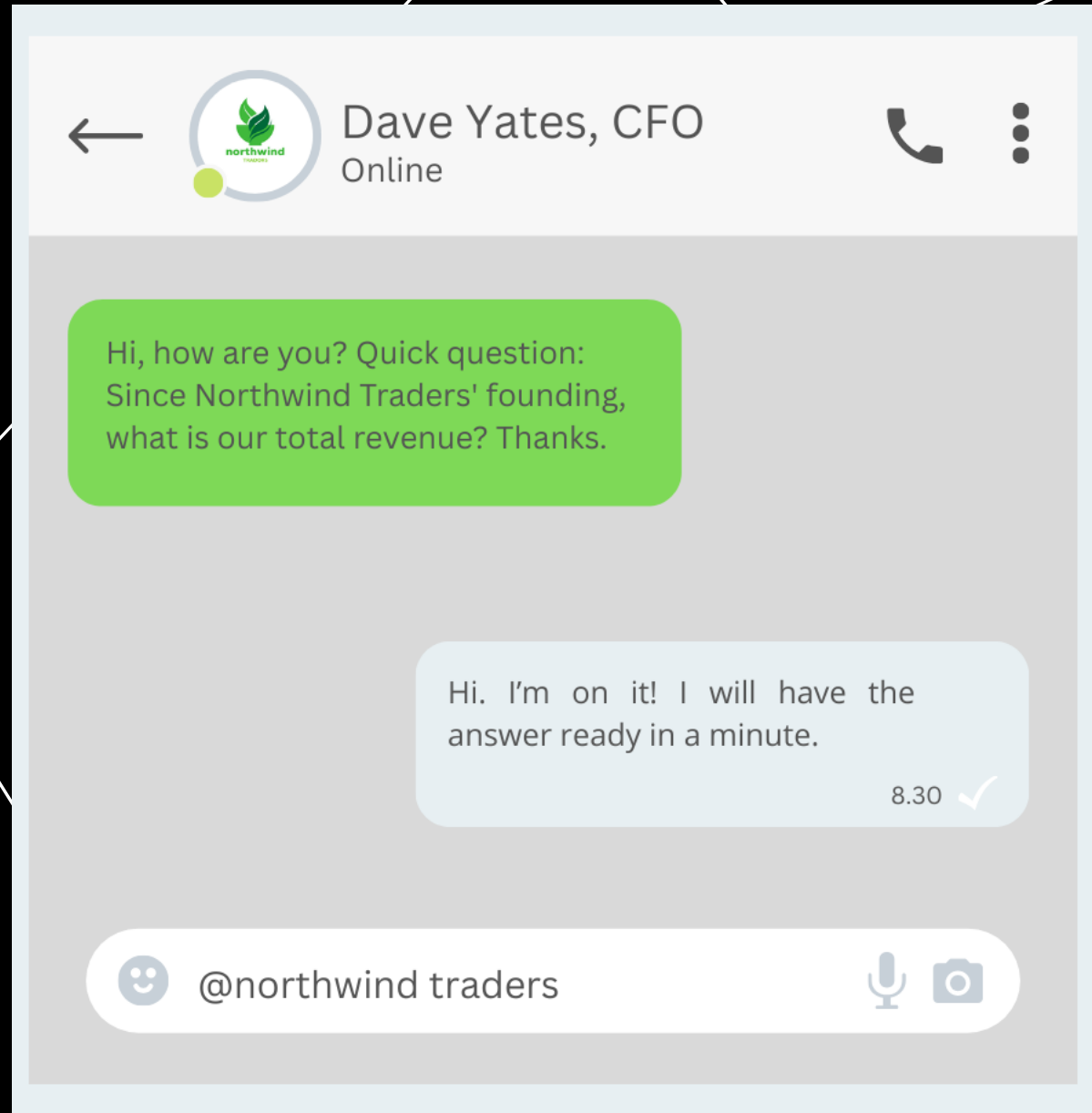
Product Name	Product ID	Year	Total Sales
Northwind Traders Coffee	43	2006	14950
Northwind Traders Beer	34	2006	6818
Northwind Traders Marmalade	20	2006	3240
Northwind Traders Mozzarella	72	2006	3132
Northwind Traders Clam Chowder	41	2006	2798.5
Northwind Traders Curry Sauce	8	2006	2600
Northwind Traders Chocolate	48	2006	2550
Northwind Traders Boysenberry Spread	6	2006	2500
Northwind Traders Crab Meat	40	2006	2208
Northwind Traders Dried Apples	51	2006	2120
Northwind Traders Ravioli	57	2006	1950
Northwind Traders Fruit Cocktail	17	2006	1560
Northwind Traders Dried Pears	7	2006	1200
Northwind Traders Cajun Seasoning	4	2006	880
Northwind Traders Chocolate Biscuits Mix	19	2006	782
Northwind Traders Green Tea	81	2006	747.5
Northwind Traders Olive Oil	5	2006	533.75
Northwind Traders Syrup	3	2006	500
Northwind Traders Long Grain Rice	52	2006	280
Northwind Traders Chai	1	2006	270
Northwind Traders Dried Plums	80	2006	262.5
Northwind Traders Scones	21	2006	200
Northwind Traders Almonds	74	2006	200

I send this view to the VP. He can clearly see that in 2006, our best-selling products were coffee (by far), followed by beer and marmalade.

He can also see that our scones and almonds performed the poorest.

COMPANY TOTAL REVENUE:

The CFO wants to know:
Since Northwind Traders founding,
what is our total revenue?



Company Total Revenue

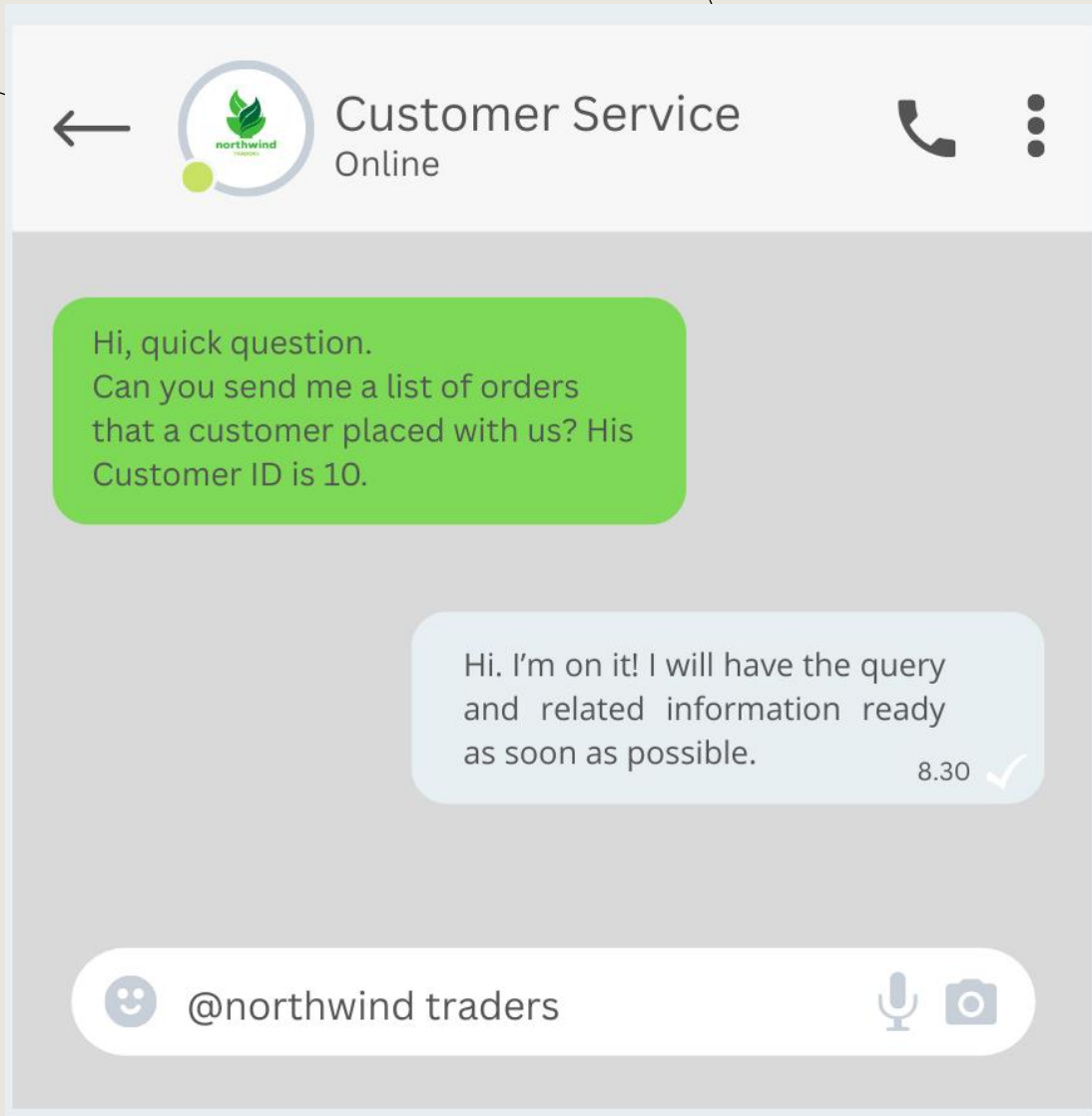
```
select sum(`Unit Price` * `Quantity`) as `Total Revenue`  
from `order details`;
```

	Total Revenue
▶	68137

I send this amount to the CFO. He can clearly see what our total revenue is since our founding.

FILTER CUSTOMER ORDERS:

A customer service rep wants to know: Can you send me a list of orders that a customer placed with us? His Customer ID is 10.



Filter Customer Orders

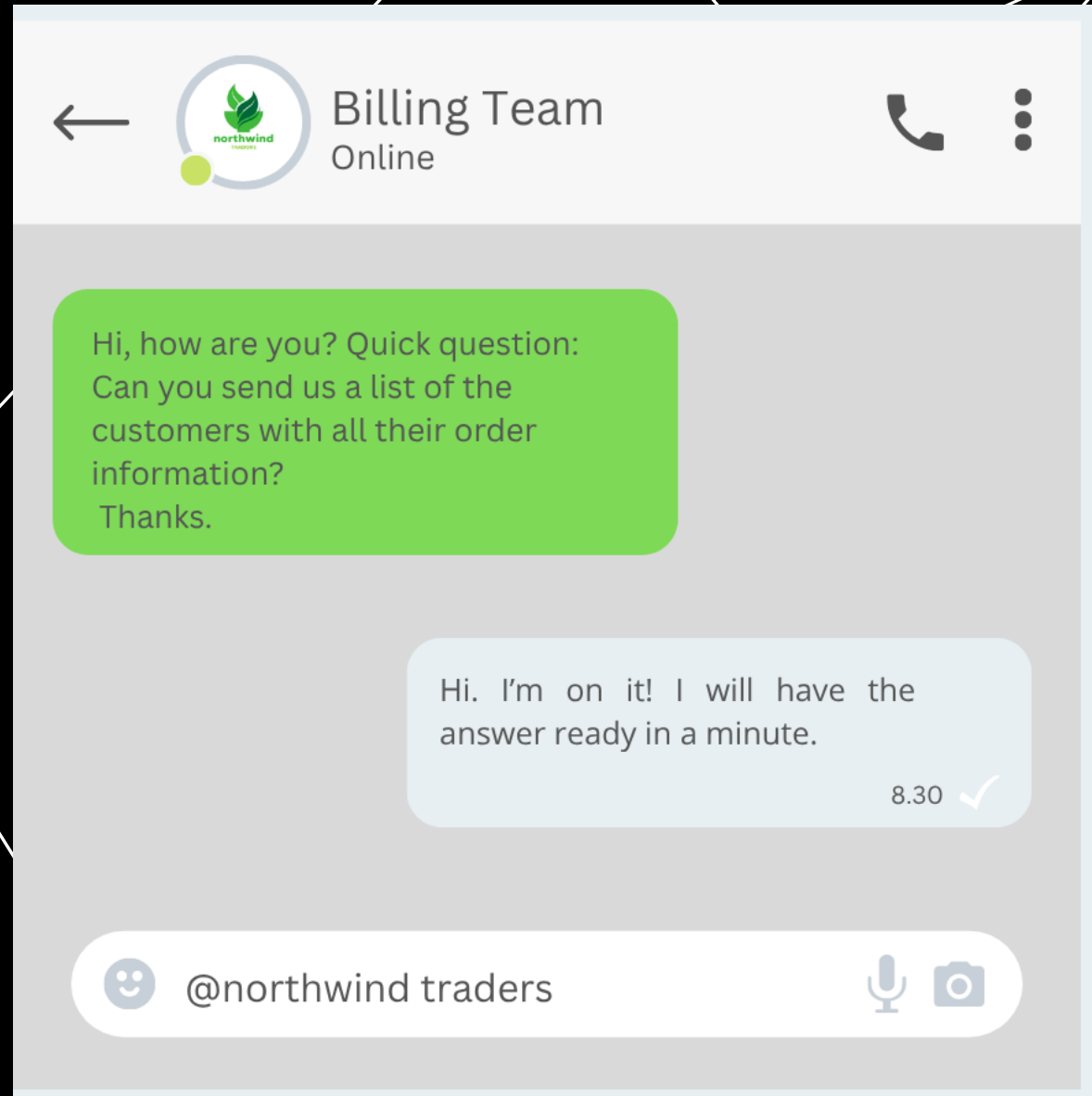
	Orders	Customer Name
►	40	Roland Wacker
	42	Roland Wacker
	67	Roland Wacker
	69	Roland Wacker

```
select `Order ID` as Orders, CONCAT(`First Name`, ' ', `Last Name`) AS `Customer Name`  
from `orders` o  
JOIN customers on customers.ID = o.Customer  
where o.Customer = 10  
GROUP BY `Order ID`, `First Name`, `Last Name`;  
;
```

I send this table to the Customer Service Rep. She can clearly see that Customer 10, Roland Wacker, has placed 4 orders with Northwind Traders. She can also see the corresponding Order IDs.

CUSTOMERS AND ORDERS:

The Billing team wants to know:
Can you send us a list of the
customers with all their order
information?

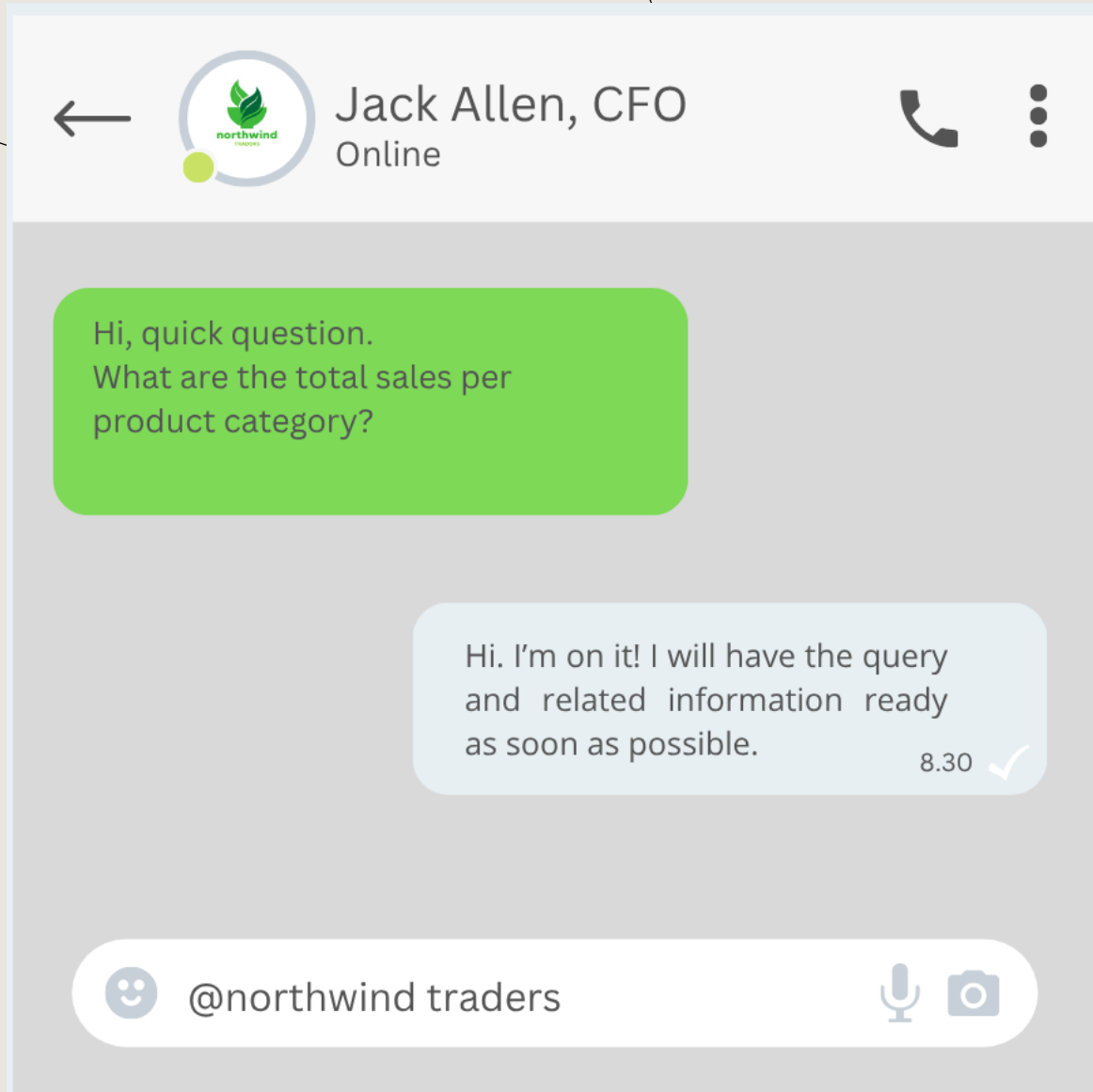


Customers and Orders

	First Name	Last Name	Order ID	Order Date
▶	Anna	Bedecs	71	5/24/2006
	Antonio	Gratacos Solsona	NULL	NULL
	Thomas	Axen	63	4/25/2006
	Thomas	Axen	36	2/23/2006
	Christina	Lee	61	4/7/2006
	Christina	Lee	58	4/22/2006
	Christina	Lee	34	2/6/2006
	Christina	Lee	31	1/20/2006
	Martin	O'Donnell	NULL	NULL
	Francisco	Pérez-Olaeta	79	6/23/2006
	Francisco	Pérez-Olaeta	74	6/8/2006
	Francisco	Pérez-Olaeta	64	5/9/2006
	Francisco	Pérez-Olaeta	56	4/3/2006
	Francisco	Pérez-Olaeta	47	4/8/2006
	Francisco	Pérez-Olaeta	37	3/6/2006
	Ming-Yang	Xie	NULL	NULL
	Elizabeth	Andersen	75	6/5/2006
	Elizabeth	Andersen	66	5/24/2006
	Elizabeth	Andersen	60	4/30/2006
	Elizabeth	Andersen	48	4/5/2006
	Elizabeth	Andersen	39	3/22/2006
	Elizabeth	Andersen	33	1/30/2006
	Sven	Mortensen	73	6/5/2006
	Sven	Mortensen	46	4/5/2006

```
select customers.`First Name`, customers.`Last Name`, orders.`Order ID`, orders.`Order Date`  
from customers  
left outer join orders  
on customers.ID= orders.customer;
```

I send this table to the Billing Department. They can clearly see the Customer name as well as the order ID and order date.



TOTAL SALES PER PRODUCT CATEGORY:

The CFO wants to know: What are the total sales per product category?

Sales per Category

	Category	count(`order details` (1).ID)
►	Beverages	13
	Dried Fruit & Nuts	9
	Baked Goods & Mixes	5
	Candy	5
	Soups	4
	Sauces	4
	Jams, Preserves	3
	Condiments	3
	Canned Meat	3
	Pasta	2
	Dairy Products	2
	Oil	1
	Canned Fruit & Ve...	1
	Grains	1

- ```
select products.Category, count(`order details`.ID)
from products
inner join `order details`
on `order details`.Product = products.`Product Name`
group by products.category;
```

I send this table to the CFO. He can clearly see how many orders we received per category.